

P3 Advanced practical skills

Practical · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Biology practical work requires clear variable definitions, biological replicates and consistent measurement. Use a microscope scale calibrated by stage micrometer; in enzyme or osmosis studies hold temperature, pH and initial material comparable. Present calculated rates with units and graphs with suitably labelled axes.

Worked example: follow the reasoning

If five 1 cm³ potato pieces gain 0.05, 0.04, 0.06, 0.05 and 0.05 g, mean mass gain is 0.05 g; replicates reveal variability.

Quick check · multiple choice

A Why use biological replicates in a practical?

[1]

A To change several conditions

B To avoid labelling axes

C To make one measurement exact

D To estimate variation

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A student studies sucrose concentration and potato mass change. Identify independent, dependent and two controlled variables.

[3]

Answers and explanation

Quick check: D. To estimate variation

Question 1 [3 marks]

Independent: sucrose concentration [1]; dependent: change or percentage change in potato mass [1]; controls: piece surface/starting mass, immersion time, temperature or solution volume [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.